

Predictors of Religiosity and Spirituality

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Abstract

The level of one's religiosity and spirituality are often associated with one's other beliefs and habits regarding issues considered related in nature to faith, including political beliefs, family relationships, church attendance and prayer habits, among others. After some preliminary exploratory data analysis to get a feel for the spread of our data, we analyzed the group differences between faith traditions, the relationship between spirituality and religiosity, behavioral and political belief predictors of religiosity and spirituality, finally culminating in the complete models that predict the spirituality and religiosity of individuals based on most available factors. The complete models were able account for more of the variance of the religiosity of an individual, with an R^2 of .71, than for the spirituality of an individual, which had an R^2 of .41.

Introduction

The dataset that we're working with comes from the PEW Research Center. The data comes from a survey of the American public about religious affiliation, conducted with the intention of tracking changes in U.S. religious affiliation. The survey was intended to get responses from as many Americans as possible, offering surveys in English and Spanish in all 50 states, with online, paper and phone format options. The data contains information about the religious, political, spiritual and social background of those surveyed. After trimming down our dataset to include only those with useful information and those without all NA values, we were left with 93 columns. Our driving question is twofold: 1) How well can we predict religiosity and spirituality based on factors including church attendance, family relationships, marriage, political opinions, religious affiliation and devotion, etc.? 2) Which factors are most effective in predicting religiosity/spirituality of individuals? Does this vary by religious affiliation? As one of the premier immigration destinations in the world, an analysis of the religious landscape in the United States provides key information on how

In our dataset, many variables were unusable due to lack of data or relevance to our research question. We narrowed our dataset down to 93 variables, of which the 28 most important were related to religious affiliation, current religion, level of religiosity and spirituality on a scale of 1-5, frequency of church service attendance, belief in God, Heaven and Hell, frequency of prayer, reading of scripture and attendance of religious services growing up, sharing beliefs with others, support of legalization of abortion, the religion of the respondents' family, marital status, support of prayer in public schools, birth decade, race and hispanic status, educational attainment, current school enrollment and employment status, immigration and voting status, political party affiliation and views, income bracket and gender. Many other variables exist and are used in this paper, but these are a sampling of many that are related to the question of religiosity and spirituality.

Exploratory Data Analysis

First, we turned the current religion variable into a factor so that we can create “bins” of how many people fall into each religious tradition.

Taking a quick

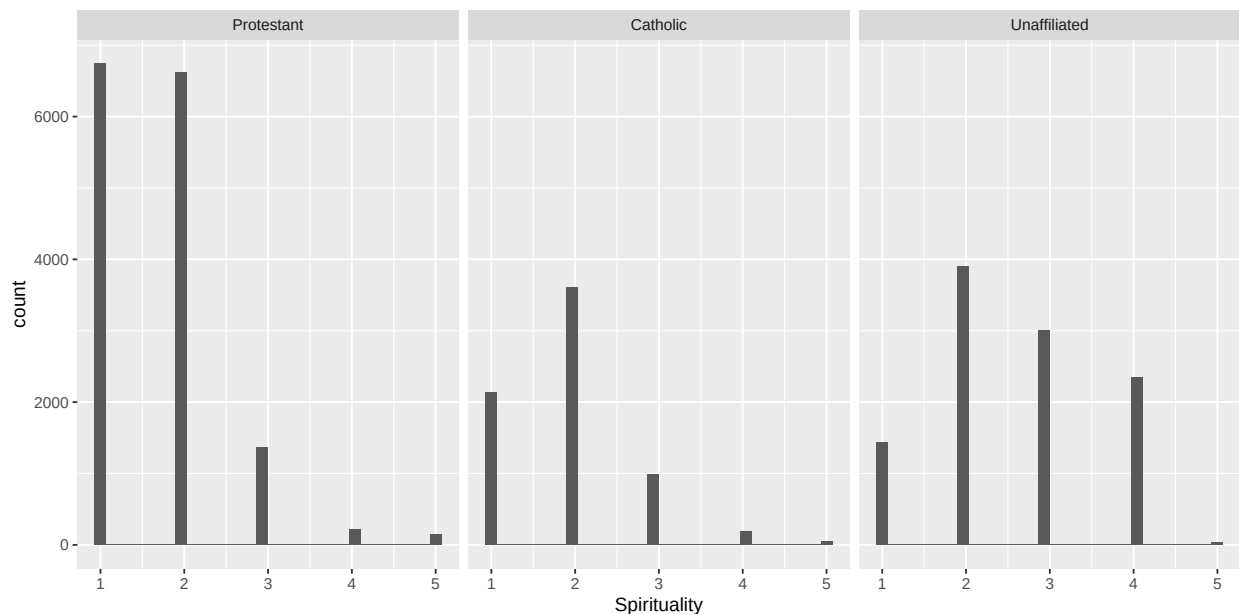
```
cor(data$SPIRPER, data$RELPER)
```

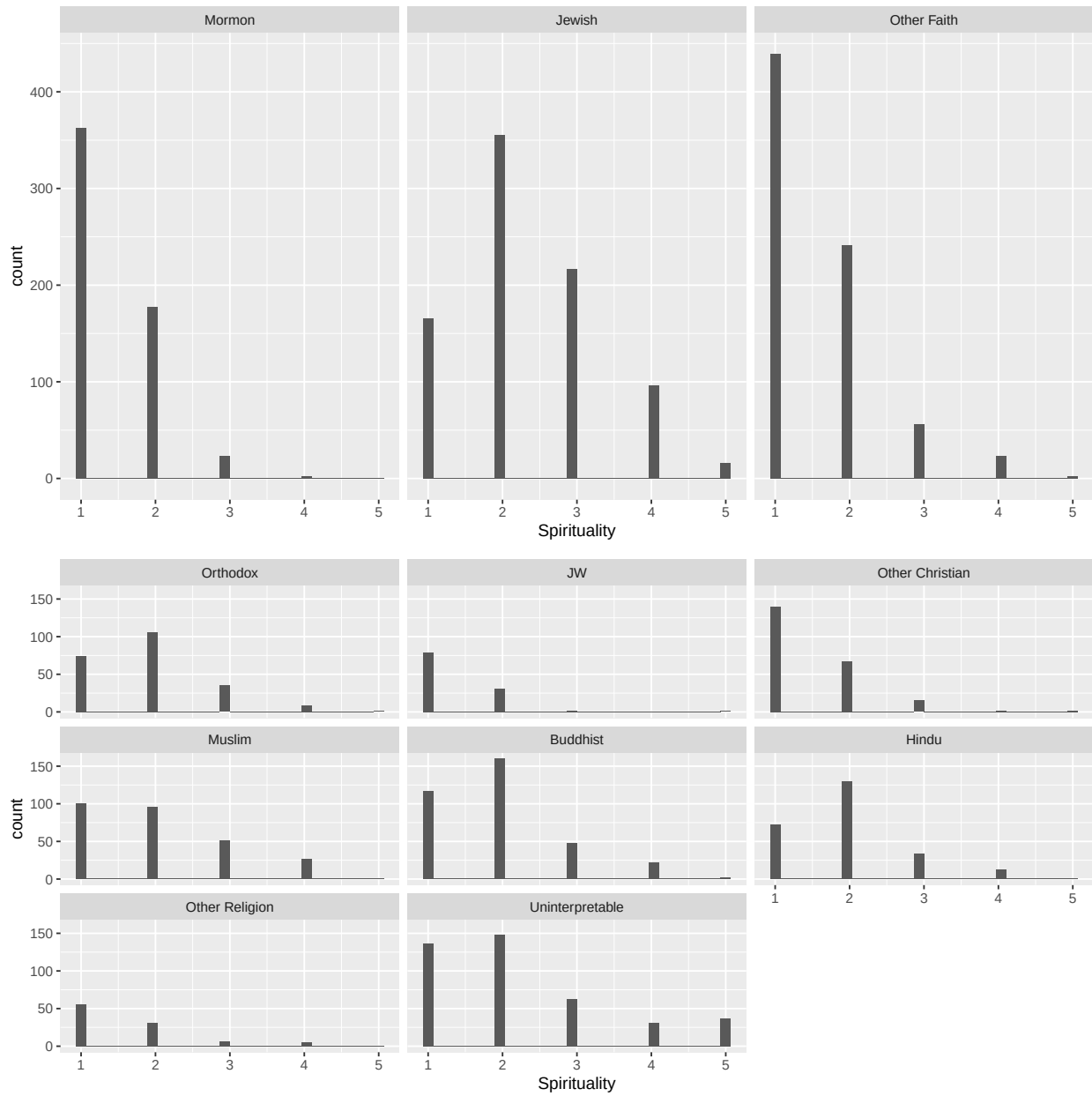
```
## [1] 0.4362961
```

From there, we were able to look at the relationship between some of the other variables and religion.

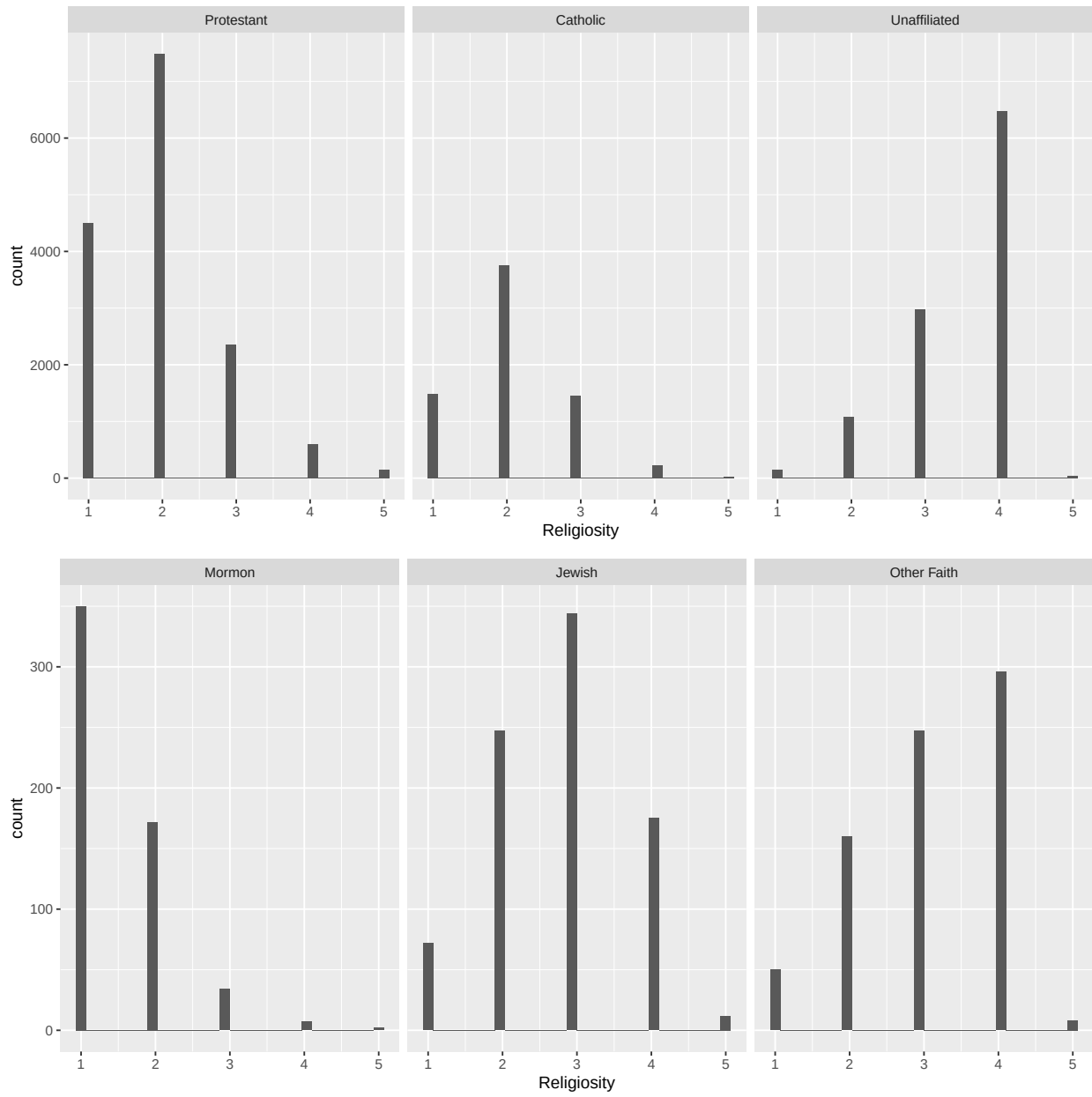
First, before trying to plot the data, we noticed that there was a much larger number of those who identified as Catholic, Protestant or Unaffiliated than anything else, so we split the histogram facet_wrap into two to account for this (then split it again to account for the much larger prevalence of Mormon, Jewish and Other Faith than all the other religious identities combined) so that the spread of each religion across the different values would be clear. This splitting up of the religious categories will prove useful throughout the duration of the EDA section.

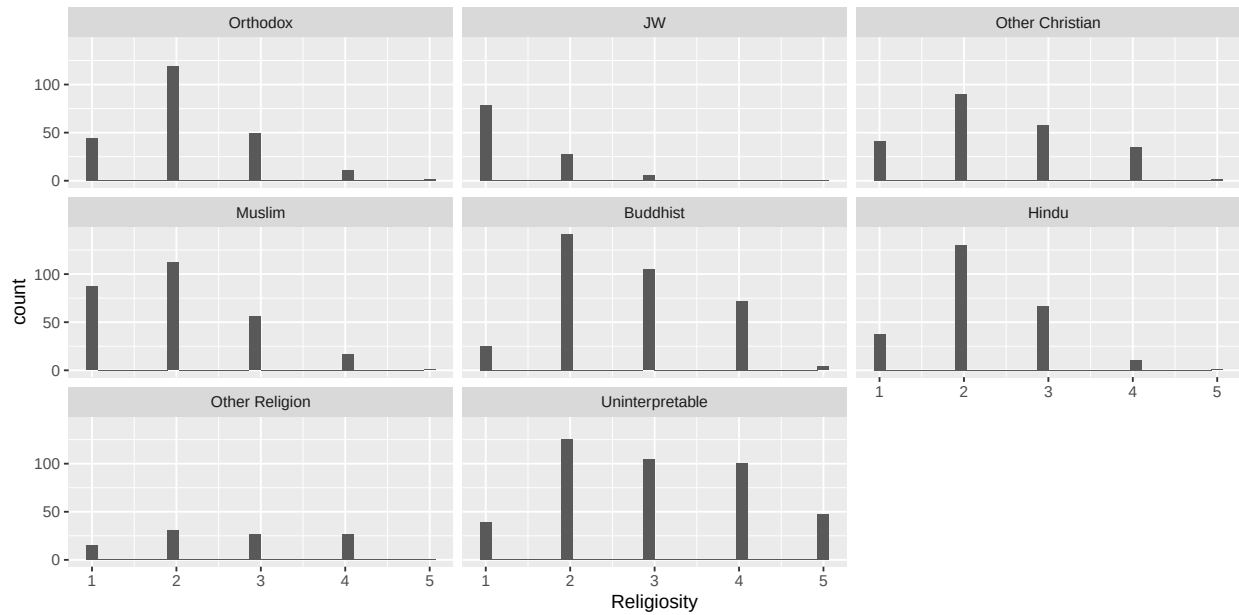
Looking at the correlation between each specific faith and personal spirituality (1=most spiritual, 4= least spiritual, 5=don't know/refused), we notice a few trends: most faiths seemed to have the largest spike at either 1 or 2. Interestingly, the Unaffiliated category also has the largest spikes around 1 and 2, and not 4. This could be that many people who identified as Unaffiliated, although they did not profess a specific religion, still considered themselves spiritual (spiritual but not religious).





The spread for religiosity is shifted more to the higher (less religious) values than spirituality: there were only two religions where the plurality (largest spike) was 1 (most religious).





Statistical Analysis

To formally investigate the relationships suggested by the exploratory data analysis, we apply several statistical methods, including analysis of variance (ANOVA), correlation analysis, simple linear regression, and multiple linear regression. These methods allow us to test for group differences, measure associations between variables, and identify key predictors of religiosity.

Section 1: Group Differences

We begin by examining whether religiosity differs across religious traditions.

Hypothesis Test:

H_0 : All religious groups have the same mean religiosity.

H_a : At least one group's mean is different.

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## CURREL      13  17394   1338.0    2247 <2e-16 ***
## Residuals 36590   21785     0.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 14 x 4
##   CURREL      mean_religiosity sd_religiosity     n
##   <fct>          <dbl>          <dbl> <int>
## 1 JW              1.38            0.631   113
## 2 Mormon           1.46            0.667   563
## 3 Protestant       1.94            0.786 14949
## 4 Muslim           2.01            0.882   272
## 5 Catholic         2.06            0.743  6928
## 6 Orthodox         2.12            0.776   223
```

## 7 Hindu	2.21	0.752	246
## 8 Other Christian	2.39	0.959	224
## 9 Other Religion	2.65	1.03	99
## 10 Buddhist	2.65	0.891	344
## 11 Uninterpretable	2.72	0.980	368
## 12 Jewish	2.74	0.884	838
## 13 Other Faith	3.05	0.933	753
## 14 Unaffiliated	3.48	0.731	10684

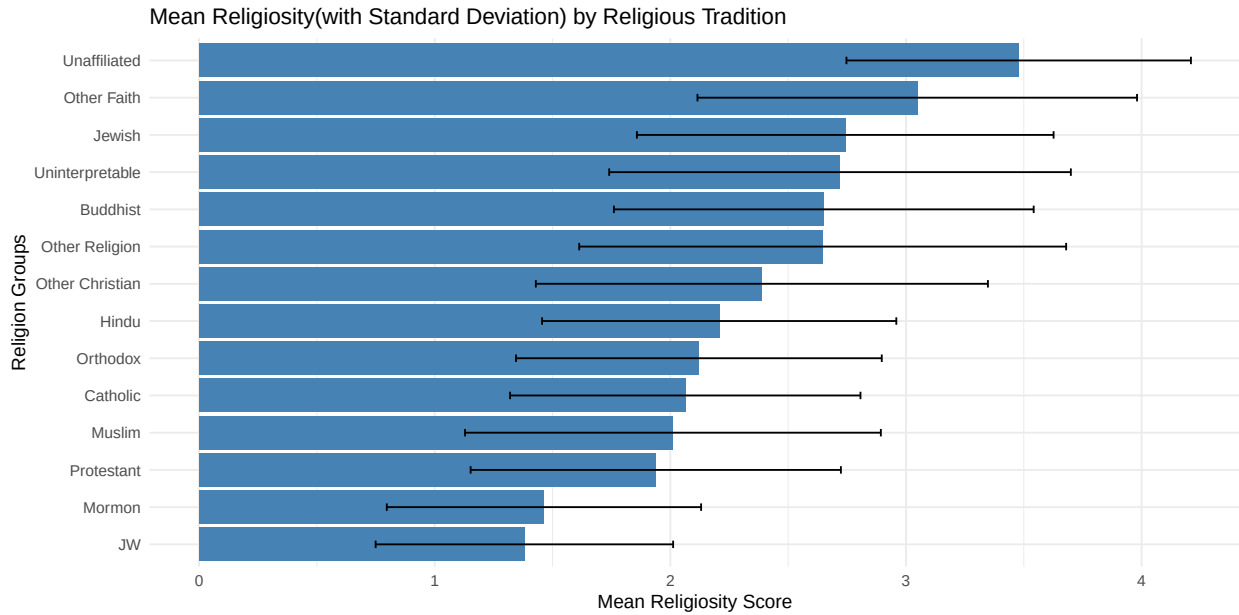
Table 1: Summary Statistics by Religious Group

Current Religion	Mean Religiosity	SD	Sample Size
JW	1.380531	0.6313051	113
Mormon	1.463588	0.6669679	563
Protestant	1.938123	0.7856999	14949
Muslim	2.011029	0.8823127	272
Catholic	2.063366	0.7434991	6928
Orthodox	2.121076	0.7761389	223
Hindu	2.207317	0.7516698	246
Other Christian	2.388393	0.9592828	224
Other Religion	2.646465	1.0332545	99
Buddhist	2.651163	0.8907355	344
Uninterpretable	2.720109	0.9795890	368
Jewish	2.742243	0.8841662	838
Other Faith	3.047809	0.9327656	753
Unaffiliated	3.478285	0.7309605	10684

The ANOVA results indicate that we reject the null hypothesis, as the p-value is less than 0.05. This suggests that religiosity differs significantly across religious traditions.

To further examine these differences, we computed the mean religiosity score for each group. Religiosity is measured on a scale from 1 (very religious) to 4 (not at all religious), excluding responses coded as 5 (don't know/refused). Because lower values indicate higher religiosity, groups with lower mean scores are more religious on average. The results show that Protestants have lower mean religiosity scores than Catholics and the unaffiliated, indicating higher average religiosity, while the unaffiliated group reports the highest mean scores, indicating the lowest levels of religiosity.

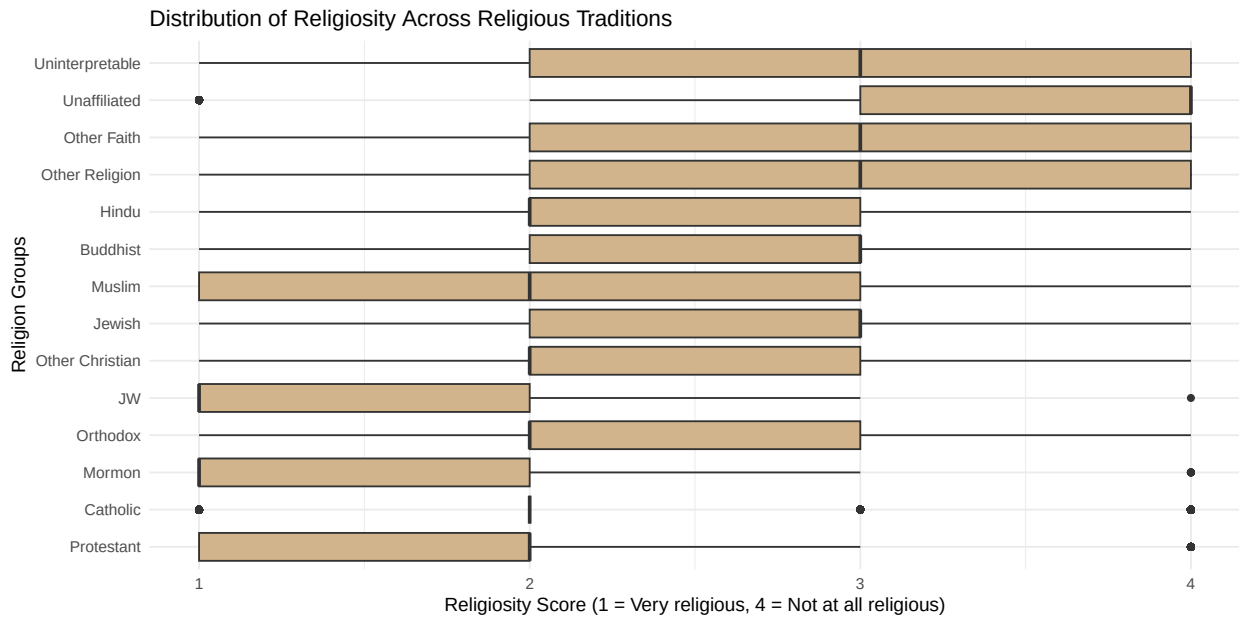
Protestants, Catholics, and the unaffiliated also have the largest sample sizes, making these estimates more reliable. The standard deviations, which range approximately from 0.5 to 1, suggest a moderate level of variation in religiosity within each group.



Check Assumptions for ANOVA: 1. Independence

2. Normality within groups

3. Common variance



Independence is assumed based on the sampling design of the dataset.

A boxplot of religiosity scores was used to examine the distribution and variability of religiosity across religious traditions. The plot shows the median, interquartile range, and potential outliers for each group.

The distributions appear reasonably symmetric within most groups, with no extreme deviations, suggesting that the assumption of approximate normality is reasonable. In addition, the spread of scores across groups is relatively similar, providing preliminary evidence that the assumption of equal variances is satisfied.

Section 2: Spirituality vs Religiosity

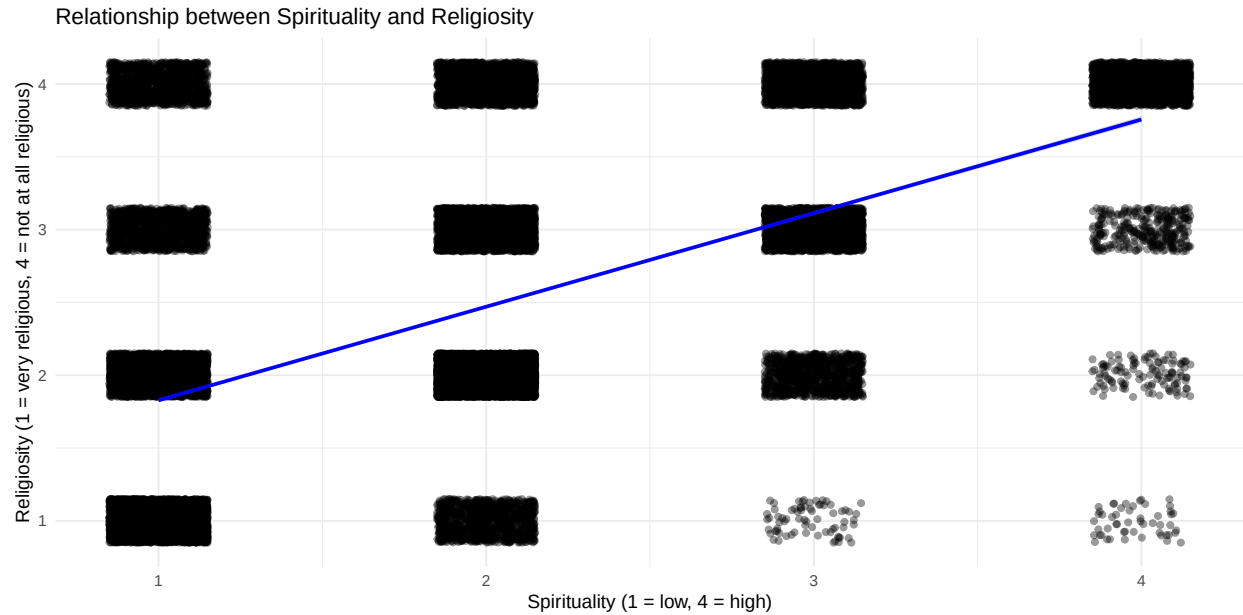
Next, we examine the relationship between the spirituality and religiosity variables, as the exploratory data analysis suggested that even unaffiliated individuals report relatively high levels of spirituality.

```
## [1] 0.561914

##
## Call:
## lm(formula = RELPER ~ SPIRPER, data = data_no5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7564 -0.4704 -0.1134  0.5296  2.1726
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.184407   0.010869   109.0   <2e-16 ***
## SPIRPER      0.642997   0.004959   129.7   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8557 on 36437 degrees of freedom
## Multiple R-squared:  0.3157, Adjusted R-squared:  0.3157
## F-statistic: 1.681e+04 on 1 and 36437 DF,  p-value: < 2.2e-16
```

A correlation analysis was conducted between religiosity and spirituality, excluding missing responses. The result is 0.562, indicating a moderate positive relationship between the two variables. This suggests that higher levels of spirituality are associated with higher religiosity scores. Given that higher religiosity scores correspond to lower levels of religiosity (1 = very religious, 4 = not at all religious), this indicates that higher spirituality is associated with lower reported religiosity.

A simple linear regression further examined this relationship. The results show that spirituality is a statistically significant predictor of religiosity ($p < 0.05$). The coefficient indicates that a one-unit increase in spirituality is associated with an increase in religiosity score. The model explains approximately 31.57% ($R^2 = 0.3157$) of the variation in religiosity, suggesting a moderate relationship.



As spirituality increases, religiosity score increased, indicating lower religiosity. A jittered scatterplot with a fitted linear regression line was used to visualize the relationship between spirituality and religiosity. Jittering was applied to help with data visualization due to the discrete nature of the 1–4 response scale.

Section 3: Behavioral Predictors

While spirituality captures internal beliefs, religiosity may also be influenced by observable behaviors such as attendance and prayer. We therefore examine whether these behavioral variables are significant predictors of religiosity.

We focused on four variables here, including frequency of religious service attendance, prayer, scripture reading, and sharing of religious views. We test whether these behavioral factors significantly predict religiosity.

Hypothesis Test:

H_0 : That none of these behaviors are associated with religiosity.

H_a : That at least one behavioral variable significantly predicts religiosity.

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = no99)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.61517 -0.40885 -0.05211  0.40467  2.84929
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.705294   0.012384  56.953  <2e-16 ***
## ATTNDPERRLS  0.215591   0.003086  69.868  <2e-16 ***
## PRAY         0.186483   0.002190  85.169  <2e-16 ***
## PRAC_B       0.063182   0.003543  17.833  <2e-16 ***
## PRAC_D      -0.004961   0.003687  -1.345    0.178
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6659 on 35842 degrees of freedom
## Multiple R-squared:  0.5856, Adjusted R-squared:  0.5855
## F-statistic: 1.266e+04 on 4 and 35842 DF,  p-value: < 2.2e-16
```

The results from the full model indicate that attendance, prayer frequency, and scripture reading are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity (higher values indicate lower religiosity), these results suggest that more frequent religious practices are associated with higher levels of religiosity (i.e., more religious behavior). In contrast, PRAC_D is not statistically significant, suggesting that sharing religious views with others does not have a measurable relationship with religiosity in this model.

The model explains approximately 58.56% ($R^2 = 0.5856$) of the variation in religiosity, indicating a relatively strong relationship between behavioral practices and religiosity.

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = protestant_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.8208 -0.4932 -0.0820  0.3761  2.6364
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.054964   0.016932  62.307 < 2e-16 ***
## ATTNDPERRLS  0.119634   0.004542  26.341 < 2e-16 ***
## PRAY         0.102805   0.004215  24.391 < 2e-16 ***
## PRAC_B       0.067325   0.005328  12.637 < 2e-16 ***
## PRAC_D       0.018922   0.005331   3.549 0.000387 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.669 on 14560 degrees of freedom
## Multiple R-squared:  0.2734, Adjusted R-squared:  0.2732
## F-statistic: 1370 on 4 and 14560 DF,  p-value: < 2.2e-16
```

The results from the Protestant model indicate that attendance (ATTNDPERRLS), prayer frequency (PRAY), scripture reading (PRAC_B), and sharing religious views (PRAC_D) are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement.

The model explains approximately 27.34% ($R^2 = 0.2734$) of the variation in religiosity, suggesting a moderate relationship between behavioral practices and religiosity within the Protestant subsample.

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = catholic_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1253 -0.3578 -0.0318  0.3484  2.8866
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.679348   0.025274  26.879 < 2e-16 ***
## ATTNDPERRLS 0.174389   0.005501  31.700 < 2e-16 ***
## PRAY        0.129122   0.004237  30.475 < 2e-16 ***
## PRAC_B      0.067755   0.006135  11.044 < 2e-16 ***
## PRAC_D      0.031407   0.007179   4.375 1.23e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5383 on 6809 degrees of freedom
## Multiple R-squared:  0.4764, Adjusted R-squared:  0.4761
## F-statistic: 1549 on 4 and 6809 DF, p-value: < 2.2e-16
```

The results from the Catholic model indicate that attendance (ATTNDPERRLS), prayer frequency (PRAY), scripture reading (PRAC_B), and sharing religious views (PRAC_D) are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement.

The model explains approximately 47.64% ($R^2 = 0.4764$) of the variation in religiosity, suggesting a relatively strong relationship between behavioral practices and religiosity within the Catholic subsample.

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = muslim_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.90472 -0.51176 -0.07219  0.41528  2.37060
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.85483    0.13110   6.521 3.54e-10 ***
## ATTNDPERRLS  0.13021    0.02980   4.369 1.79e-05 ***
## PRAY        0.16148    0.02480   6.512 3.72e-10 ***
## PRAC_B      0.06696    0.03438   1.948  0.0525 .
## PRAC_D      0.02527    0.03519   0.718  0.4732
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6871 on 264 degrees of freedom
## Multiple R-squared:  0.4063, Adjusted R-squared:  0.3973
## F-statistic: 45.16 on 4 and 264 DF, p-value: < 2.2e-16
```

The results from the Muslim model indicate that attendance (ATTNDPERRLS), prayer frequency (PRAY), and scripture reading (PRAC_B) are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement. In contrast, PRAC_D is not statistically significant, suggesting that sharing religious views with others does not have a measurable relationship with religiosity in this model.

The model explains approximately 40.63% ($R^2 = 0.4063$) of the variation in religiosity, suggesting a moderate to relatively strong relationship between behavioral practices and religiosity within the Muslim subsample.

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = unaffil_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8749 -0.3659  0.1711  0.3101  2.5644
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.818845   0.044116  18.561 < 2e-16 ***
## ATTNDPERRLS  0.227620   0.007980  28.524 < 2e-16 ***
## PRAY         0.161981   0.003658  44.286 < 2e-16 ***
## PRAC_B       0.125079   0.008699  14.379 < 2e-16 ***
## PRAC_D      -0.022988   0.006884  -3.339 0.000843 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5709 on 10515 degrees of freedom
## Multiple R-squared:  0.3884, Adjusted R-squared:  0.3881
## F-statistic: 1669 on 4 and 10515 DF, p-value: < 2.2e-16
```

The results from the Unaffiliated model indicate that some behavioral variables are statistically significant predictors of religiosity ($p < 0.05$). Given the coding of religiosity, where higher values indicate lower levels of religiosity, these results suggest that more frequent religious practices are associated with lower religiosity scores, indicating higher levels of religious engagement.

The model explains approximately 38.84% ($R^2 = 0.3884$) of the variation in religiosity, suggesting a moderate relationship between behavioral practices and religiosity within the Unaffiliated subsample.

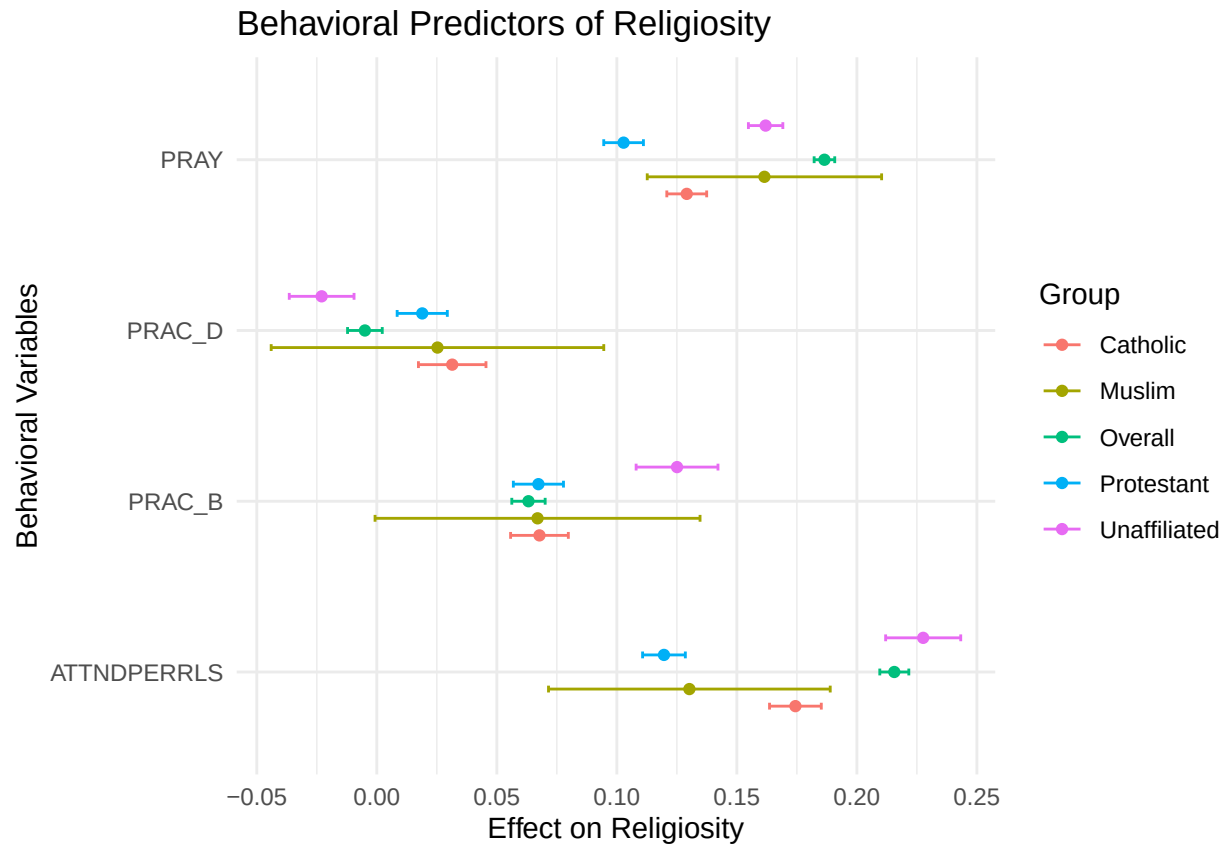
```
models <- list(
  Overall = model2,
  Protestant = model_protestant,
  Catholic = model_catholic,
  Muslim = model_muslim,
  Unaffiliated = model_unaffil
)

coef_df <- bind_rows(lapply(names(models), function(name) {
  tidy(models[[name]], conf.int = TRUE) |>
    filter(term != "(Intercept)") |>
    mutate(Group = name)
}))

ggplot(coef_df, aes(x = estimate, y = term, color = Group)) +
  geom_point(position = position_dodge(width = 0.5)) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high),
    position = position_dodge(width = 0.5), height = 0.2) +
  labs(
    title = "Behavioral Predictors of Religiosity",
    x = "Effect on Religiosity",
    y = "Behavioral Variables"
  ) +
  theme_minimal()
```

```
## Warning: 'geom_errorbarh()' was deprecated in ggplot2 4.0.0.
## i Please use the 'orientation' argument of 'geom_errorbar()' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## 'height' was translated to 'width'.
```



Section 4: Political Predictors of Religiosity

```
##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1085 -0.6519 -0.1085  0.8915  3.7058
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.6830172   0.0156302  235.634   <2e-16 ***
## ABRTLGL      -0.2550419   0.0065717  -38.809   <2e-16 ***
## GAYMARR      -0.3135173   0.0077475  -40.467   <2e-16 ***
## QB2A         -0.0049037   0.0166560   -0.294    0.7684
```

```
## CHNG_C      -0.0010580  0.0004984  -2.123   0.0338 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.921 on 35221 degrees of freedom
## Multiple R-squared:  0.2378, Adjusted R-squared:  0.2377
## F-statistic: 2748 on 4 and 35221 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = SPIRPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3393 -0.6044 -0.1779  0.6618  3.5559
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.683e+00  1.527e-02 175.670 < 2e-16 ***
## ABRTLGL      -1.603e-01  6.420e-03 -24.966 < 2e-16 ***
## GAYMARR       -1.145e-01  7.569e-03 -15.133 < 2e-16 ***
## QB2A          -6.954e-02  1.627e-02  -4.274 1.93e-05 ***
## CHNG_C         1.156e-05  4.870e-04   0.024  0.981
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8998 on 35221 degrees of freedom
## Multiple R-squared:  0.08104, Adjusted R-squared:  0.08094
## F-statistic: 776.5 on 4 and 35221 DF,  p-value: < 2.2e-16
```

It would appear that removal of the 99s which represent those who did not answer made a substantial difference on the fit of the model. Furthermore, support for gay marriage predicts religiosity well, as does support for transgender individuals. However, acceptance of homosexuality is an incredibly poor predictor of religiosity in the body public.

Alternatively, support for abortion, homosexuality, and gay marriage predicts spirituality but support for transgender individuals does not predict spirituality.

```
##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = catholic_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.5059 -0.4432 -0.1394  0.4941  3.5568
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.843525  0.032614  87.186 <2e-16 ***
## ABRTLGL      -0.225040  0.011091 -20.290 <2e-16 ***
## GAYMARR       -0.137512  0.013897  -9.895 <2e-16 ***
## QB2A          0.026926  0.027925   0.964  0.335
## CHNG_C       -0.001988  0.011860  -0.168  0.867
## ---
```

```

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7036 on 6512 degrees of freedom
## Multiple R-squared:  0.1519, Adjusted R-squared:  0.1514
## F-statistic: 291.6 on 4 and 6512 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = protestant_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3114 -0.6037 -0.0915  0.3070  3.3963
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.579250   0.026161  98.591 < 2e-16 ***
## ABRTLGL      -0.089316   0.008836 -10.108 < 2e-16 ***
## GAYMARR      -0.124174   0.010350 -11.998 < 2e-16 ***
## QB2A         -0.064033   0.020881  -3.067  0.00217 **
## CHNG_C        0.003236   0.010320   0.314  0.75384
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7991 on 14034 degrees of freedom
## Multiple R-squared:  0.07049, Adjusted R-squared:  0.07023
## F-statistic: 266.1 on 4 and 14034 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = unaffiliated_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7160 -0.5340  0.2840  0.4403  2.2287
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.159420   0.029526 140.874 <2e-16 ***
## ABRTLGL      -0.156293   0.010387 -15.046 <2e-16 ***
## GAYMARR      -0.172493   0.013189 -13.079 <2e-16 ***
## QB2A         -0.023650   0.030521  -0.775   0.438
## CHNG_C       -0.091006   0.009786  -9.300 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6885 on 10420 degrees of freedom
## Multiple R-squared:  0.1083, Adjusted R-squared:  0.108
## F-statistic: 316.4 on 4 and 10420 DF,  p-value: < 2.2e-16

```

While support for abortion, same sex marriage, and homosexuality is a statistically significant predictor of religiosity in Catholics and people who are unaffiliated, support for these causes is not statistically significant in predicting religiosity in Protestants. Interestingly, greater acceptance of transgender individuals cannot be used to predict religiosity in people who are unaffiliated.

```
##
## Call:
## lm(formula = RELPER ~ CHNG_A + CHNG_B + DIVRELPOP + DIVRACPOP +
##      QB2C, data = catholic_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.28838 -0.09125 -0.08980  0.90463  3.14626
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.0916628  0.0108237 193.249  <2e-16 ***
## CHNG_A       0.0007251  0.0009820   0.738  0.4603
## CHNG_B      -0.0020007  0.0015930  -1.256  0.2092
## DIVRELPOP    -0.0023070  0.0013777  -1.675  0.0941 .
## DIVRACPOP     0.0020643  0.0015143   1.363  0.1729
## QB2C        -0.0001745  0.0008096  -0.216  0.8294
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7637 on 6511 degrees of freedom
## Multiple R-squared:  0.000831, Adjusted R-squared:  6.374e-05
## F-statistic: 1.083 on 5 and 6511 DF, p-value: 0.3674

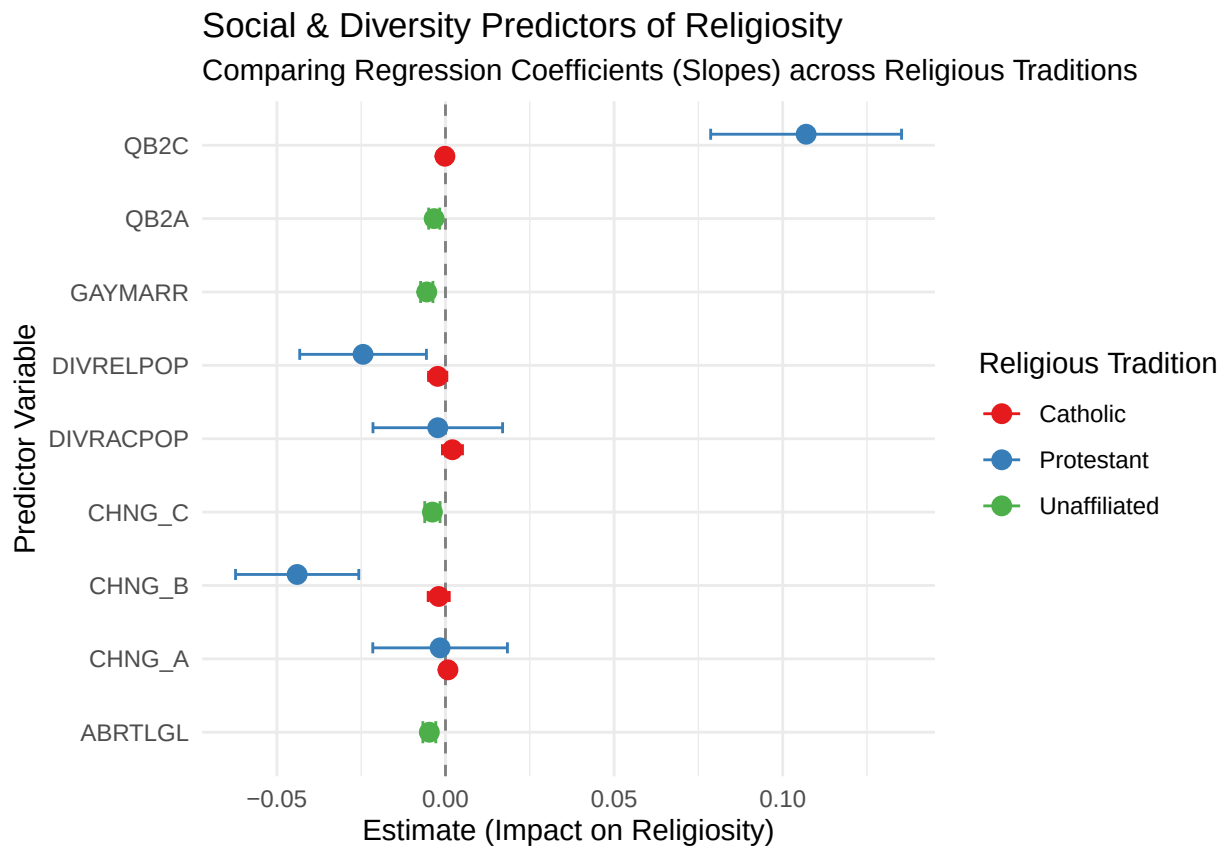
##
## Call:
## lm(formula = RELPER ~ CHNG_A + CHNG_B + DIVRELPOP + DIVRACPOP +
##      QB2C, data = protestant_pol2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0617 -0.8802 -0.0097  0.1592  3.1882
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.920164  0.037825  50.764  < 2e-16 ***
## CHNG_A      -0.001597  0.010184  -0.157  0.8754
## CHNG_B      -0.043990  0.009323  -4.718 2.40e-06 ***
## DIVRELPOP    -0.024439  0.009585  -2.550  0.0108 *
## DIVRACPOP    -0.002287  0.009806  -0.233  0.8156
## QB2C         0.106948  0.014440   7.406 1.37e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8257 on 14284 degrees of freedom
## Multiple R-squared:  0.008846, Adjusted R-squared:  0.008499
## F-statistic: 25.5 on 5 and 14284 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = unaffiliated_pol2)
##
## Residuals:
```



```
## -2.5074 -0.4996 0.4926 0.4974 2.0436
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.5249508  0.0078804 447.307 < 2e-16 ***
## ABRTLGL      -0.0047903  0.0009737  -4.920 8.81e-07 ***
## GAYMARR       -0.0055488  0.0009298  -5.968 2.48e-09 ***
## QB2A          -0.0033568  0.0008324  -4.033 5.56e-05 ***
## CHNG_C        -0.0038806  0.0011546  -3.361 0.000779 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.728 on 10428 degrees of freedom
## Multiple R-squared:  0.01229,    Adjusted R-squared:  0.01191
## F-statistic: 32.43 on 4 and 10428 DF,  p-value: < 2.2e-16

## 'height' was translated to 'width'.
```



Section 5: Comprehensive Model

Finally, we construct a comprehensive model using 27 variables that incorporates behavioral measures, belief variables, and religious affiliation to determine which factors are most important in predicting religiosity.

```
##
```

```
## Call:
## lm(formula = RELPER ~ RELTRAD + CURREL + ATTNDPERRLS + GOD +
##     HVN + HLL + PRAY + PRAC_D + PRAC_B + ABRTLGL + FRMREL + CHATTEND +
##     MARITAL + SCPRY2 + BIRTHDECADE + HISP + RACECMB + EDUCREC +
##     E2 + EMPLSIT + USGEN + REG + PARTY + IDEO + INC_SDT1 + GENDER,
##     data = data_new)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9463 -0.3907  0.0026  0.3380  3.8385
##
## Coefficients: (13 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.1546751   0.0572450    2.702 0.006897 **
## RELTRADMainline -0.0920514   0.0131457   -7.002 2.57e-12 ***
## RELTRADBlackprot -0.0686408   0.0204429   -3.358 0.000787 ***
## RELTRADCatholic -0.1558562   0.0147685  -10.553 < 2e-16 ***
## RELTRADMormon   -0.2731317   0.0383345   -7.125 1.07e-12 ***
## RELTRADOrthodox -0.1147863   0.0642830   -1.786 0.074168 .
## RELTRADJW       -0.4067587   0.0946345   -4.298 1.73e-05 ***
## RELTRADOther Christian 0.0196182   0.0491388    0.399 0.689719
## RELTRADJewish    -0.1143146   0.0415891   -2.749 0.005988 **
## RELTRADMuslim    -0.1192571   0.0631132   -1.890 0.058825 .
## RELTRADBuddhist  -0.1163810   0.0407735   -2.854 0.004316 **
## RELTRADHindu     -0.2139961   0.0826045   -2.591 0.009585 **
## RELTRADOther Religion -0.0532060   0.0740842   -0.718 0.472651
## RELTRADOther Faith  0.1928352   0.0273614    7.048 1.86e-12 ***
## RELTRADUnaffiliated 0.3105669   0.0148299   20.942 < 2e-16 ***
## RELTRADUninterpretable 0.1453885   0.0475682    3.056 0.002242 **
## CURRELCatholic      NA         NA         NA      NA
## CURRELMormon        NA         NA         NA      NA
## CURRELOrthodox      NA         NA         NA      NA
## CURRELJW            NA         NA         NA      NA
## CURRELOther Christian NA         NA         NA      NA
## CURRELJewish        NA         NA         NA      NA
## CURRELMuslim        NA         NA         NA      NA
## CURRELBuddhist      NA         NA         NA      NA
## CURRELHindu         NA         NA         NA      NA
## CURRELOther Religion NA         NA         NA      NA
## CURRELOther Faith   NA         NA         NA      NA
## CURRELUnaffiliated  NA         NA         NA      NA
## CURRELUninterpretable NA         NA         NA      NA
## ATTNDPERRLS        0.1435577   0.0034371   41.767 < 2e-16 ***
## GOD                0.2670802   0.0128124   20.845 < 2e-16 ***
## HVN                0.2244932   0.0116475   19.274 < 2e-16 ***
## HLL                0.0841924   0.0105365    7.991 1.39e-15 ***
## PRAY               0.0800703   0.0026828   29.846 < 2e-16 ***
## PRAC_D             0.0194993   0.0038410    5.077 3.87e-07 ***
## PRAC_B             0.0701480   0.0038813   18.073 < 2e-16 ***
## ABRTLGL            -0.0205819   0.0050303   -4.092 4.30e-05 ***
## FRMRELCatholic     0.0527613   0.0110445    4.777 1.79e-06 ***
## FRMRELMormon       0.0144971   0.0329209    0.440 0.659678
## FRMRELOrthodox     -0.0495179   0.0586126   -0.845 0.398211
## FRMRELJW           0.0648343   0.0617742    1.050 0.293940
```

```

## FRMRELOther Christian    0.1331098    0.0461763    2.883 0.003946 **
## FRMRELJewish            0.0177724    0.0379591    0.468 0.639646
## FRMRELMuslim            -0.0000515    0.0623826   -0.001 0.999341
## FRMRELBuddhist          -0.0889790    0.0409087   -2.175 0.029633 *
## FRMRELHindu             -0.0391475    0.0772467   -0.507 0.612309
## FRMRELOther Religion    -0.0291839    0.1025704   -0.285 0.776009
## FRMRELOther Faith       -0.0456088    0.0672209   -0.678 0.497466
## FRMRELUnaaffiliated     0.0081807    0.0145273    0.563 0.573353
## FRMRELUinterpretatable  -0.0413862    0.0584715   -0.708 0.479074
## CHATTEND                 0.0153417    0.0030007    5.113 3.20e-07 ***
## MARITAL                 -0.0009687    0.0020042   -0.483 0.628836
## SCPRY2                   0.0493508    0.0045118   10.938 < 2e-16 ***
## BIRTHDECADE              0.0013130    0.0029816    0.440 0.659681
## HISP                    -0.0079116    0.0136409   -0.580 0.561925
## RACECMB                  0.0038982    0.0030393    1.283 0.199649
## EDUCREC                  0.0133080    0.0041480    3.208 0.001337 **
## E2                       0.0081239    0.0091951    0.884 0.376971
## EMPLSIT                  -0.0137532    0.0027189   -5.058 4.26e-07 ***
## USGEN                    0.0021144    0.0063429    0.333 0.738868
## REG                      0.0056203    0.0114479    0.491 0.623466
## PARTY                    0.0151994    0.0040954    3.711 0.000207 ***
## IDEO                     0.0296278    0.0043501    6.811 9.89e-12 ***
## INC_SDT1                 0.0054207    0.0018968    2.858 0.004269 **
## GENDER                   0.0256205    0.0072889    3.515 0.000440 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6074 on 29113 degrees of freedom
## (7743 observations deleted due to missingness)
## Multiple R-squared:  0.6645, Adjusted R-squared:  0.6639
## F-statistic: 1131 on 51 and 29113 DF, p-value: < 2.2e-16

```

With an R^2 of 0.71, we have pretty good confidence that this model can predict religiosity in individuals. An R^2 of 0.71 tells us that 71% of variance can be explained by the model and the variables that we presented it with. The results for spirituality were quite a bit lower, however, with an R^2 of just .41.

```

##
## Call:
## lm(formula = SPIRPER ~ RELTRAD + CURREL + ATTNDPERRLS + GOD +
##     HVN + HLL + PRAY + PRAC_D + PRAC_B + ABRTLGL + FRMREL + CHATTEND +
##     MARITAL + SCPRY2 + BIRTHDECADE + HISP + RACECMB + EDUCREC +
##     E2 + EMPLSIT + USGEN + REG + PARTY + IDEO + INC_SDT1 + GENDER,
##     data = data_new)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.3573 -0.4403 -0.0784  0.4716  3.8317
##
## Coefficients: (13 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.5643517  0.0658849   8.566 < 2e-16 ***
## RELTRADMainline 0.0349419  0.0151297   2.309 0.020923 *
## RELTRADBlackprot 0.0226823  0.0235282   0.964 0.335032
## RELTRADCatholic 0.0780179  0.0169974   4.590 4.45e-06 ***

```

## RELTRADMormon	-0.0984053	0.0441202	-2.230	0.025729	*
## RELTRADOrthodox	-0.1896945	0.0739851	-2.564	0.010354	*
## RELTRADJW	0.0292758	0.1089175	0.269	0.788094	
## RELTRADOther Christian	-0.3418322	0.0565553	-6.044	1.52e-09	***
## RELTRADJewish	-0.0287206	0.0478661	-0.600	0.548498	
## RELTRADMuslim	0.2131685	0.0726388	2.935	0.003342	**
## RELTRADBuddhist	-0.4690225	0.0469273	-9.995	< 2e-16	***
## RELTRADHindu	-0.1408043	0.0950718	-1.481	0.138609	
## RELTRADOther Religion	-0.5108922	0.0852656	-5.992	2.10e-09	***
## RELTRADOther Faith	-0.6044468	0.0314910	-19.194	< 2e-16	***
## RELTRADUnaffiliated	-0.0211013	0.0170682	-1.236	0.216359	
## RELTRADUninterpretable	-0.2576474	0.0547475	-4.706	2.54e-06	***
## CURRELCatholic	NA	NA	NA	NA	
## CURELMormon	NA	NA	NA	NA	
## CURRELOrthodox	NA	NA	NA	NA	
## CURELJW	NA	NA	NA	NA	
## CURRELOther Christian	NA	NA	NA	NA	
## CURELJewish	NA	NA	NA	NA	
## CURELMuslim	NA	NA	NA	NA	
## CURELBuddhist	NA	NA	NA	NA	
## CURELHindu	NA	NA	NA	NA	
## CURRELOther Religion	NA	NA	NA	NA	
## CURRELOther Faith	NA	NA	NA	NA	
## CURELUnaffiliated	NA	NA	NA	NA	
## CURELUninterpretable	NA	NA	NA	NA	
## ATTNDPERRLS	0.0299768	0.0039558	7.578	3.62e-14	***
## GOD	0.5882078	0.0147462	39.889	< 2e-16	***
## HVN	-0.0409528	0.0134055	-3.055	0.002253	**
## HLL	-0.0126176	0.0121267	-1.040	0.298127	
## PRAY	0.1301858	0.0030877	42.163	< 2e-16	***
## PRAC_D	0.0661743	0.0044207	14.969	< 2e-16	***
## PRAC_B	0.0502192	0.0044671	11.242	< 2e-16	***
## ABRTLGL	0.0056728	0.0057895	0.980	0.327175	
## FRMRELCatholic	-0.0270337	0.0127114	-2.127	0.033452	*
## FRMRELMormon	-0.0370594	0.0378895	-0.978	0.328037	
## FRMRELOrthodox	0.2136180	0.0674589	3.167	0.001544	**
## FRMRELJW	-0.0552682	0.0710976	-0.777	0.436955	
## FRMRELOther Christian	-0.0053787	0.0531456	-0.101	0.919387	
## FRMRELJewish	0.1059830	0.0436881	2.426	0.015277	*
## FRMRELMuslim	-0.0017946	0.0717979	-0.025	0.980059	
## FRMRELBuddhist	0.2239042	0.0470829	4.756	1.99e-06	***
## FRMRELHindu	0.0947644	0.0889053	1.066	0.286477	
## FRMRELOther Religion	0.2560152	0.1180511	2.169	0.030115	*
## FRMRELOther Faith	0.0665290	0.0773664	0.860	0.389839	
## FRMRELUnaffiliated	0.0596997	0.0167199	3.571	0.000357	***
## FRMRELUninterpretable	0.1475065	0.0672965	2.192	0.028395	*
## CHATTEND	0.0149232	0.0034536	4.321	1.56e-05	***
## MARITAL	0.0003277	0.0023066	0.142	0.887019	
## SCPRY2	0.0185967	0.0051928	3.581	0.000343	***
## BIRTHDECADE	0.0166531	0.0034316	4.853	1.22e-06	***
## HISP	0.0427771	0.0156997	2.725	0.006440	**
## RACECMB	-0.0021368	0.0034981	-0.611	0.541292	
## EDUCREC	-0.0389498	0.0047741	-8.159	3.52e-16	***
## E2	-0.0133410	0.0105829	-1.261	0.207456	

```
## EMPLSIT          0.0077070  0.0031293   2.463 0.013789 *
## USGEN            -0.0365648  0.0073002  -5.009 5.51e-07 ***
## REG              0.0448067  0.0131757   3.401 0.000673 ***
## PARTY            -0.0157780  0.0047135  -3.347 0.000817 ***
## IDEO             -0.0231026  0.0050067  -4.614 3.96e-06 ***
## INC_SDT1         0.0071798  0.0021831   3.289 0.001007 **
## GENDER           -0.1211495  0.0083890 -14.441 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6991 on 29113 degrees of freedom
## (7743 observations deleted due to missingness)
## Multiple R-squared:  0.4346, Adjusted R-squared:  0.4336
## F-statistic: 438.7 on 51 and 29113 DF,  p-value: < 2.2e-16
```

Check for Assumptions of Regression: 1. Linearity 2. Nearly normal residuals 3. Constant variability

```
library(modelr)
```

```
## Warning: package 'modelr' was built under R version 4.5.3
```

```
##
```

```
## Attaching package: 'modelr'
```

```
## The following object is masked from 'package:broom':
```

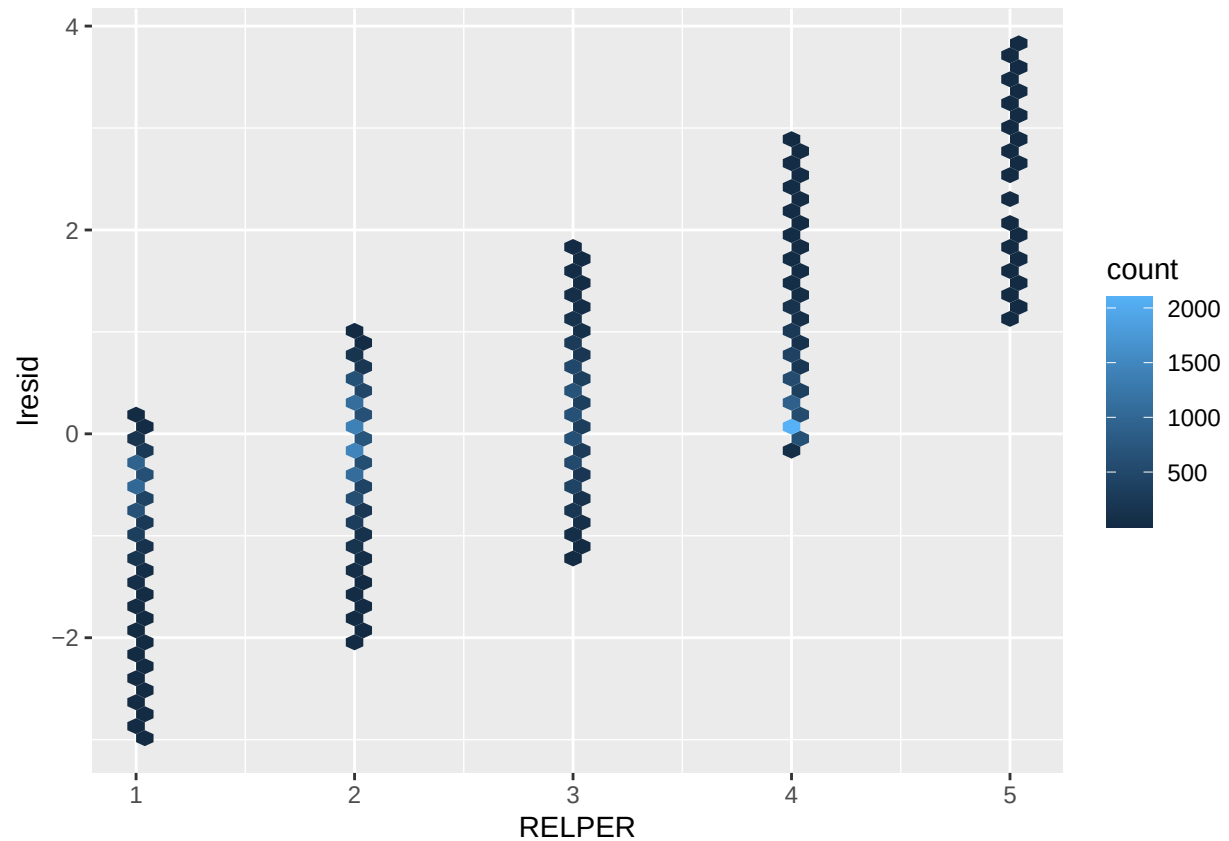
```
##
```

```
## bootstrap
```

```
residuals_plot_rel <- data_new %>%
  add_residuals(final_model_rel, "lresid")

ggplot(residuals_plot_rel, aes(RELPER, lresid)) +
  geom_hex(bins = 50)
```

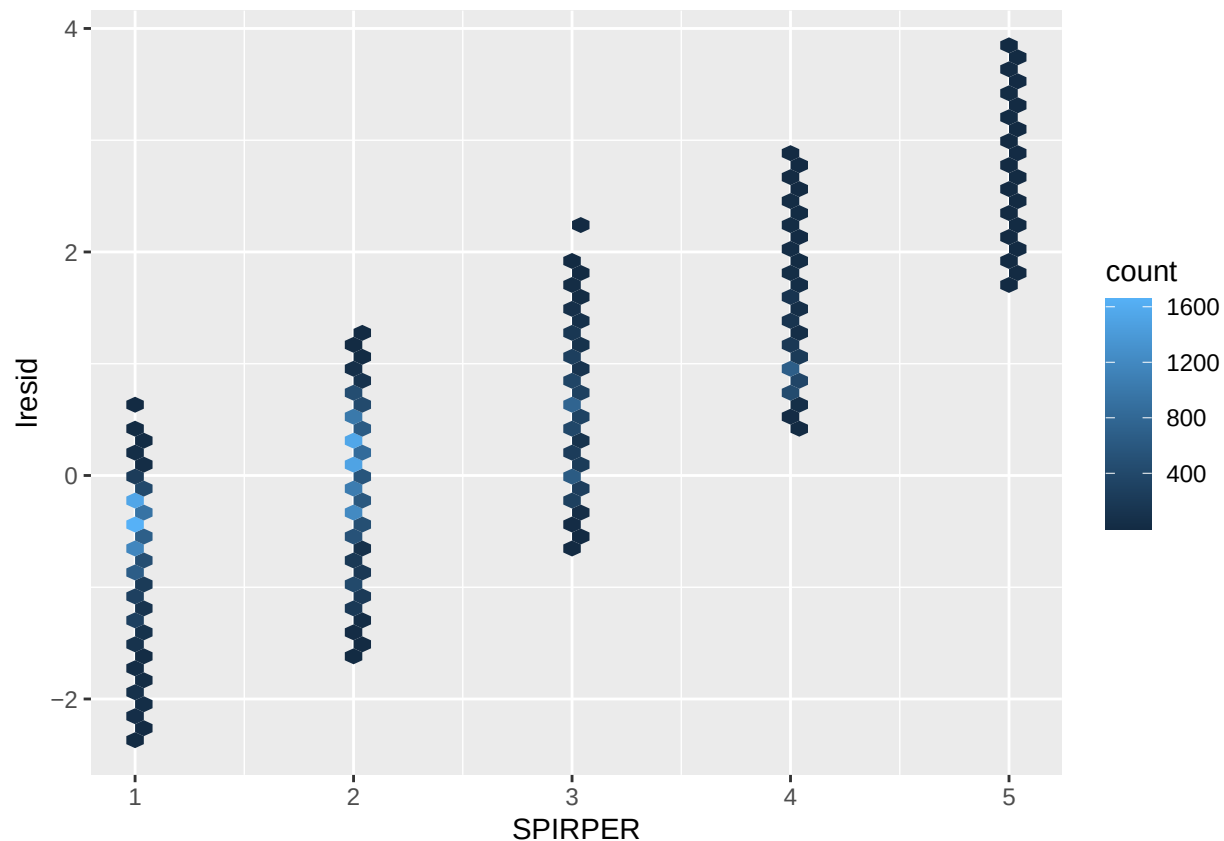
```
## Warning: Removed 7743 rows containing non-finite outside the scale range
## ('stat_binhex()').
```



```
residuals_plot_spir <- data_new %>%
  add_residuals(final_model_spir, "lresid")

ggplot(residuals_plot_spir, aes(SPIRPER, lresid)) +
  geom_hex(bins = 50)
```

```
## Warning: Removed 7743 rows containing non-finite outside the scale range
## ('stat_binhex()').
```



```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + GOD + HVN + HLL + PRAY +
##      PRAC_D + PRAC_B + ABRTLGL + SCPRY2 + EDUCREC + EMPLSIT +
##      IDEO + INC_SDT1, data = data_new)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8760 -0.4040  0.0103  0.3406  3.9674
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0070719  0.0290511  -0.243  0.80767
## ATTNDPERRLS  0.1777559  0.0033242  53.473 < 2e-16 ***
## GOD          0.3692938  0.0126142  29.276 < 2e-16 ***
## HVN          0.2919996  0.0114533  25.495 < 2e-16 ***
## HLL          0.0890206  0.0104308   8.534 < 2e-16 ***
## PRAY         0.0939623  0.0026357  35.650 < 2e-16 ***
## PRAC_D       0.0180148  0.0038435   4.687 2.78e-06 ***
## PRAC_B       0.0622569  0.0037580  16.566 < 2e-16 ***
## ABRTLGL     -0.0151776  0.0049411  -3.072 0.00213 **
## SCPRY2       0.0610627  0.0045068  13.549 < 2e-16 ***
## EDUCREC      0.0017535  0.0041380   0.424 0.67175
## EMPLSIT     -0.0236684  0.0021494 -11.012 < 2e-16 ***
## IDEO         0.0399762  0.0042709   9.360 < 2e-16 ***
## INC_SDT1    -0.0007988  0.0017555  -0.455 0.64908
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6297 on 30460 degrees of freedom
## (6434 observations deleted due to missingness)
## Multiple R-squared:  0.6393, Adjusted R-squared:  0.6391
## F-statistic: 4153 on 13 and 30460 DF, p-value: < 2.2e-16
```

Looking at the p-values for the religiosity model, there were quite a few variables that were not impactful in predicting religiosity. These included FRMREL, RELTRAD, CURREL, CHATTEND, MARITAL, BIRTHDECADE, HISP, RACECMB, E2, USGEN, REG, PARTY, and GENDER. When these variables were removed, the R^2 value decreased to .64.

Conclusions

Across both the full sample and some religious subsamples, behavioral factors—including attendance, prayer frequency, and scripture reading—consistently emerge as significant predictors of religiosity. These findings suggest that religiosity is strongly shaped by observable religious practices, rather than being solely an internal belief system.

However, the strength of these relationships varies across religious traditions. The Catholic and Muslim subsamples show relatively stronger explanatory power, while the Protestant model shows a weaker relationship between behaviors and religiosity. The Unaffiliated group also demonstrates that behavioral practices remain relevant even in the absence of formal religious affiliation.

Overall, the results indicate that religiosity is best understood as a behaviorally structured phenomenon, but the degree to which behavior predicts religiosity differs across religious traditions. This suggests that religious tradition plays a moderating role in shaping how behaviors translate into levels of religiosity.

Similarly, depending on the religious tradition, certain political beliefs seemed to show a moderate correlation with religiosity and spirituality.

For the full models, religiosity was much easier to predict based on the behavioral, political and religious beliefs and factors than spirituality. One reason that this may be the case is potentially due to religious affiliation (with higher religiosity being already more rare) being associated with a specific set of beliefs that one subscribes to. On the other hand, one can claim to be spiritual to a certain extent, but this term is more vague in what it describes. Many classify themselves as “spiritual but not religious,” a category that denotes vague spiritual feelings without adherence to a specific religion. It is important to note that the analogous “religious but not spiritual” designation is almost far less common than its counterpart and mostly nonsensical.

Source: <https://www.pewresearch.org/dataset/2023-24-religious-landscape-study-rls-dataset/>